

SITEAI EQUIPMENT TRACKER: YOLOV8 DEPLOYMENT

Autonomous Object Detection for AECO Heavy Equipment

Master's in Artificial Intelligence for Architecture & Construction

Block 2 | Key Areas of AI Applied to the AECO Sector

Module 4

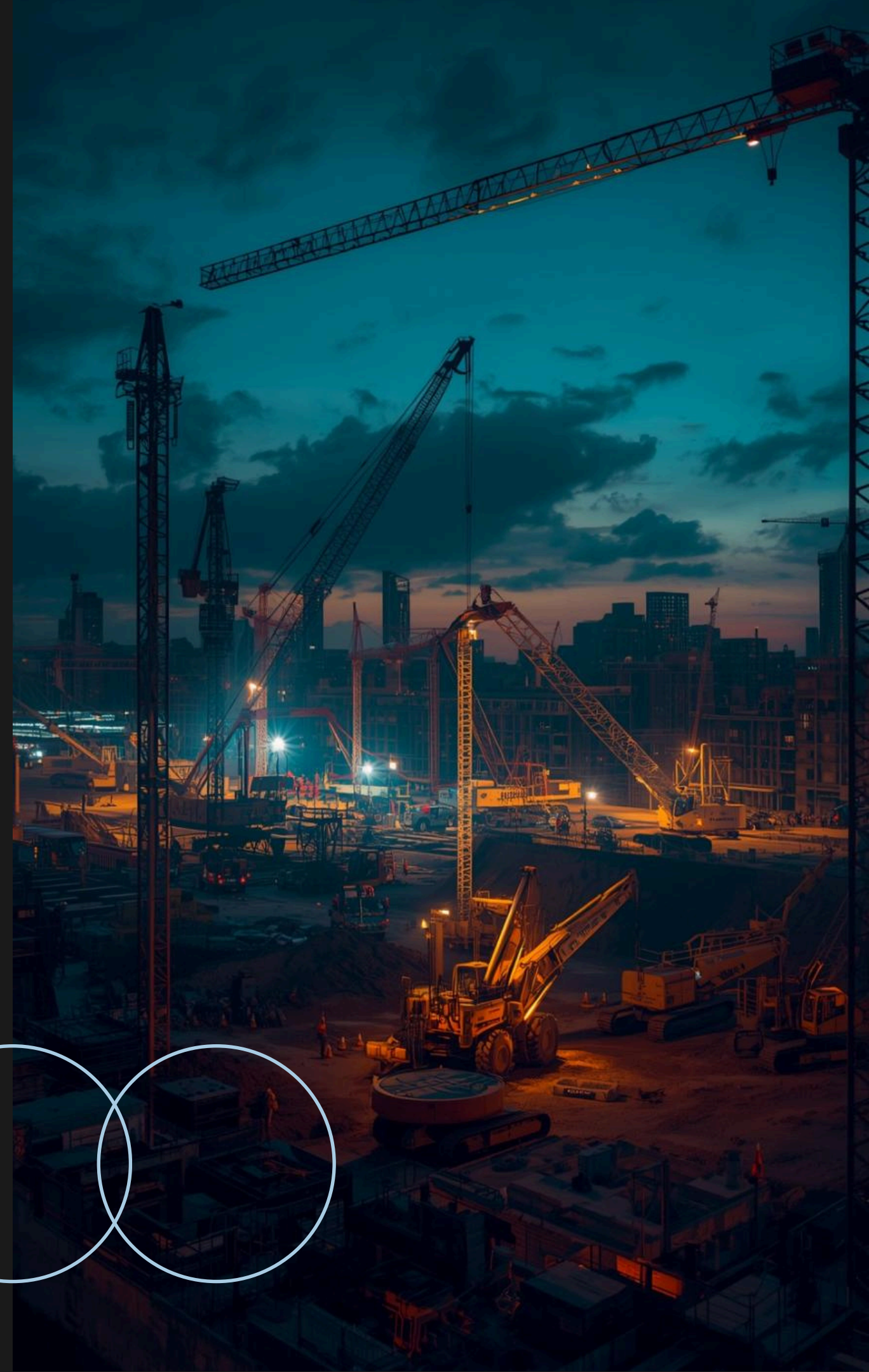
Programming and Artificial Intelligence for the AECO Sector

M4U3 | Team Assignment

GROUP MEMBERS:

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Vagia Dimara, Juan Pablo Garcia



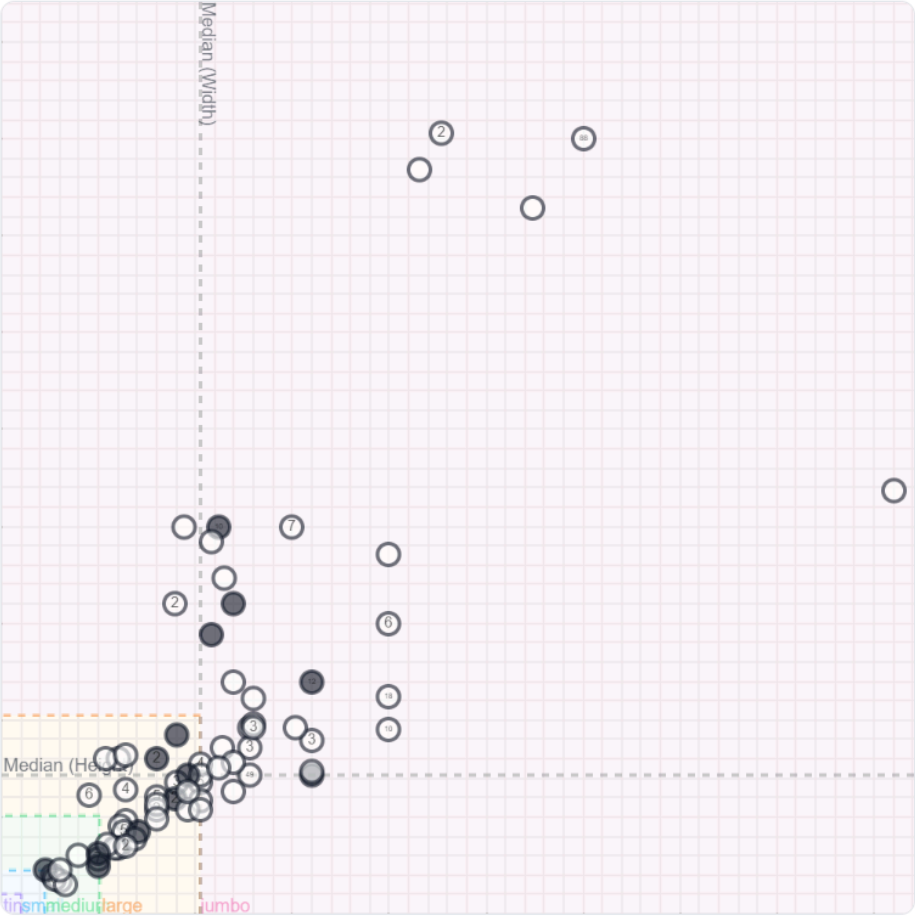
Stage 1: Dataset Analytics & Health

Dimension Insights ③

Overview of the sizes and aspect ratios of the images in your dataset.

Class ▼ Tags ▼

The dashed lines represent the **median width** (1024 px) and **median height** (720 px) of images in your dataset.



Sizes 495 total images

jumbo 244 49.3%

large 213 43.0%

medium 38 7.7%

Aspect Ratios

very wide 130 26.3%

wide 231 46.7%

tall 125 25.3%

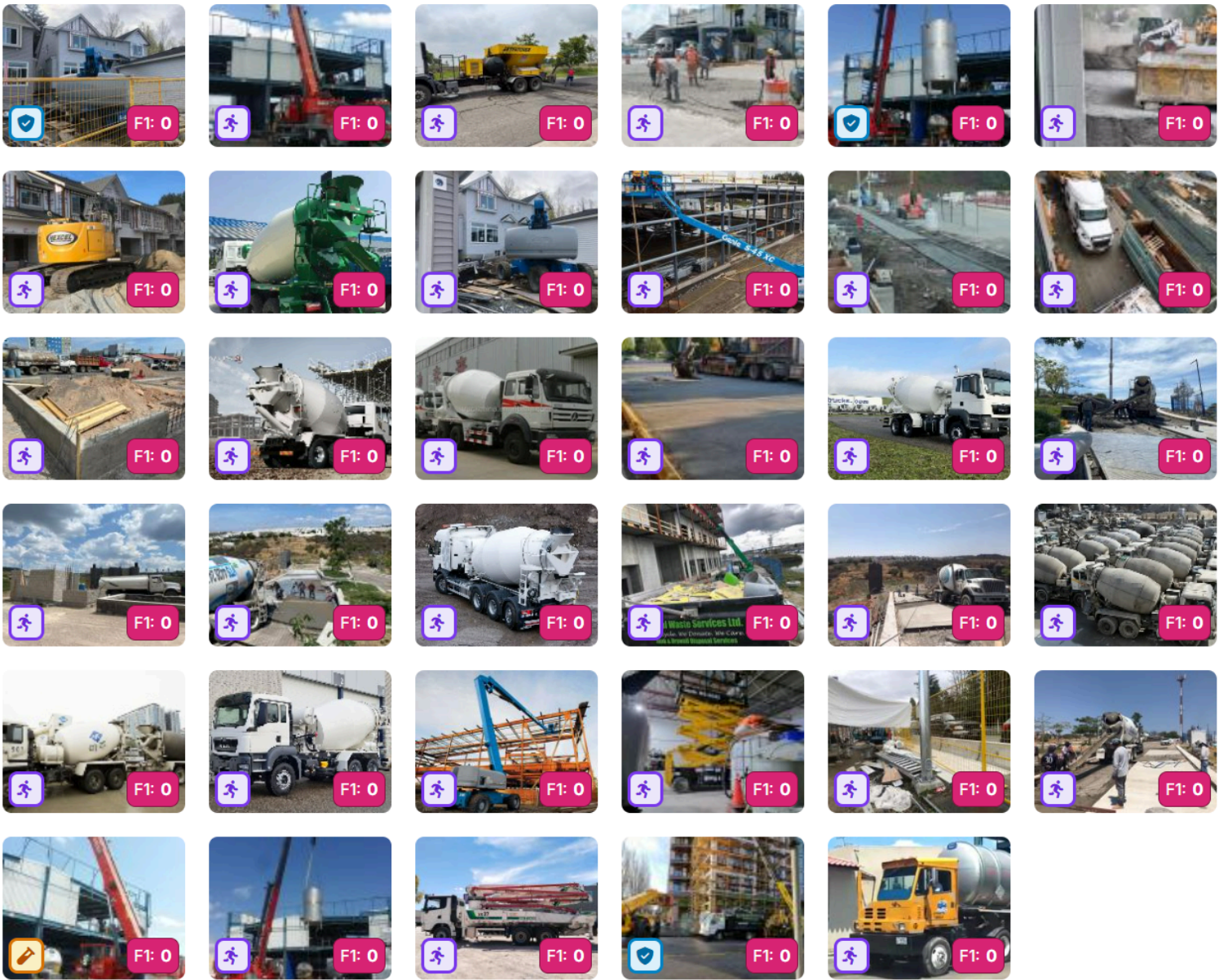
very tall 1 0.2%

square 8 1.6%

Images

Reset

Add Tags to 35 Images



The dataset

The SiteAI Equipment Tracker dataset contains 581 source images annotated across 7 construction equipment classes: boom_lift, dump_truck, excavator, loader, mixer_truck, roller, and tower_crane. Through augmentation (rotation $\pm 3^\circ$, brightness $\pm 15\%$, exposure $\pm 10\%$) at 3 outputs per training example, the dataset expands to 1,509 total images. The data is split 80/20 into training and validation sets, preprocessed with auto-orient and resized to 512x512 pixels. The augmentation strategy is intentionally conservative to maintain realistic construction site conditions while helping address class imbalance across the 7 equipment categories.

Research Repository & Source Code:

<https://github.com/Vagiadim/SiteAI-equipment-tracker->
[Includes: Training Notebooks,](#)
[Dataset v13 Analytics, and](#)
[Model Weights, README file](#)

Figure 1: Roboflow dataset annotation (screenshot of one of the stages of Roboflow experiments

Stage 2: Reproducible Pipeline

Roboflow link: <https://universe.roboflow.com/juanp-garcia/siteai-equipment-tracker/dataset/18>

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REPRODUCIBILITY SUMMARY

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Date/Time:

2026-02-27 17:33:38

Ultralytics ver:

8.4.18

PyTorch ver:

2.10.0+cu128

GPU:

Tesla T4

Model:

YOLOv8n (Nano)

Dataset:

SiteAI-equipment-tracker (Team7)

Split:

80/20 (train/val)

Epochs:

50

--- Model Comparison ---

YOLOv8n:

3M params, 8.1 GFLOPs

YOLOv8s:

11.2M params, 28.6 GFLOPs

Image size:

512

Batch:

auto

Patience:

15 (early stopping)

=====

Figure 2: Google Colab notebook Cell 12

Benchmarking & Convergence

• Extensive Exploration: Before finalizing our custom Colab YOLOv8 models, we benchmarked native cloud architectures including YOLOv11 and Roboflow 3.0.

• Baseline Validation: The RF-DETR (Small) model established an exceptionally strong early baseline (89.2% mAP@50), validating the dataset's overall learning potential.

Training Configuration & Hyperparameters

• Standardized Pipeline: All data was pre-processed to a 512x512 resolution using a strict 80/20 train/validation split to ensure honest evaluation.

• Optimization: Models were trained for a maximum of 50 epochs with an early stopping patience of 15 to prevent overfitting.

• Architecture Comparison: We run a direct comparison between the lightweight YOLOv8n (3M parameters) and the heavier YOLOv8s (11.2M parameters) to find the best balance for edge deployment.

Versions

dataset_v3_yolov8_512...

v18

1509

Small

siteai-equipment-tracker/10

JuanPablo Garcia

Version Name:

dataset_v3_yolov8_512_train_FINAL

✓

Source Images

Images: 581

Classes: 7

Unannotated: 0

✓

Train/Test Split

Training Set: 464 images

Validation Set: 59 images

Testing Set: 58 images

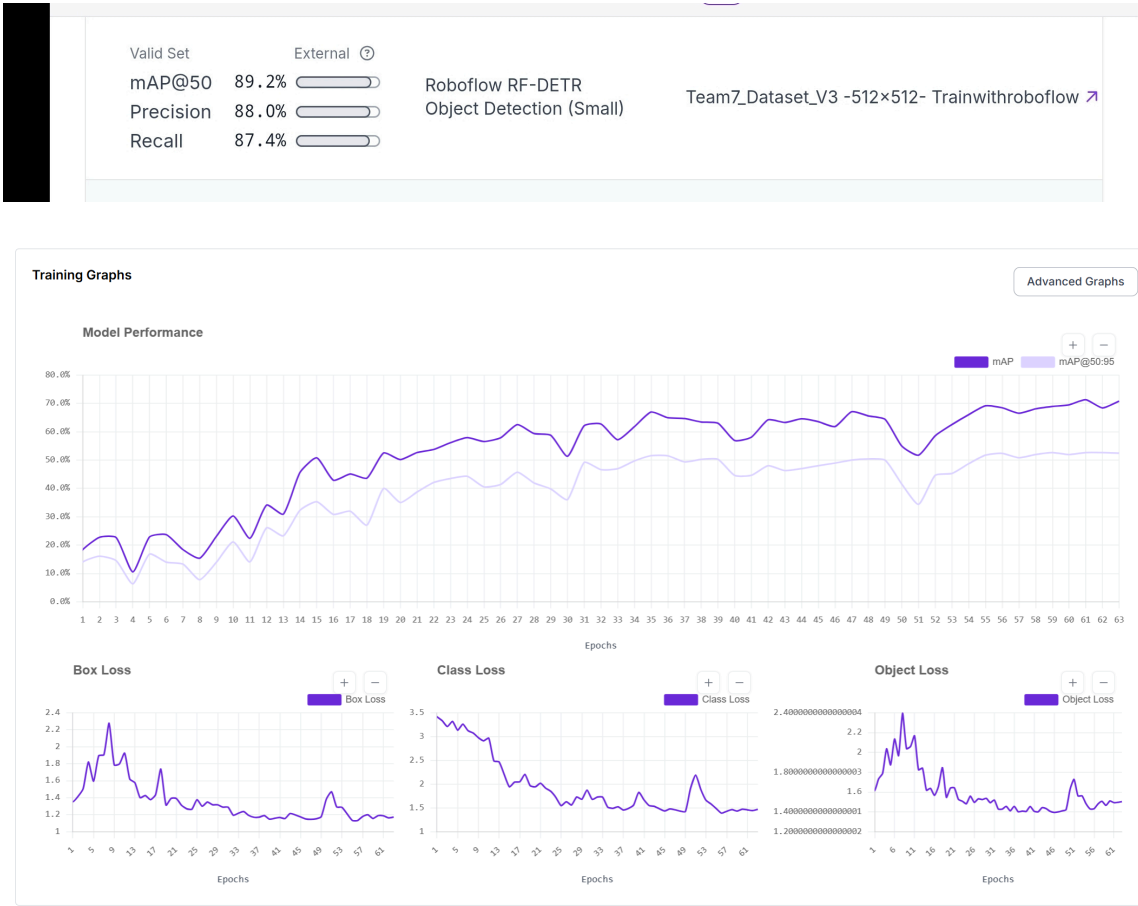
✓

Preprocessing

Auto-Orient: Applied

Resize: Stretch to 512×512

Figure 3: Roboflow training screenshots



v18 dataset_v3_yolov8_512_train_FINAL

Generated on Feb 24, 2026

Download Dataset

Popular Download Formats

YOLO26

YOLOv12

YOLOv11

YOLOv9

YOLOv8

YOLOv5

YOLOv7

COCO JSON

YOLO Darknet

Pascal VOC XML

TFRecord

PaliGemma

CreateML JSON

Other Formats

1509 Total Images

View All Images →

Dataset Split

TRAIN SET

1392 Images

92%

VALID SET

59 Images

4%

TEST SET

58 Images

4%

Preprocessing

Auto-Orient: Applied

Resize: Stretch to 512x512

Augmentations

Outputs per training example: 3

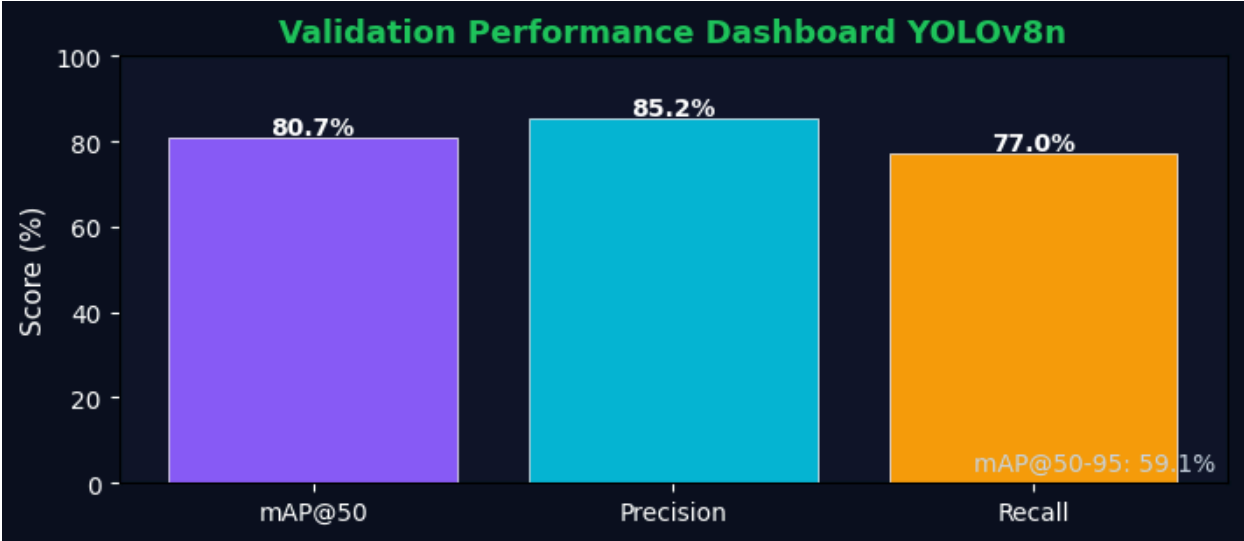
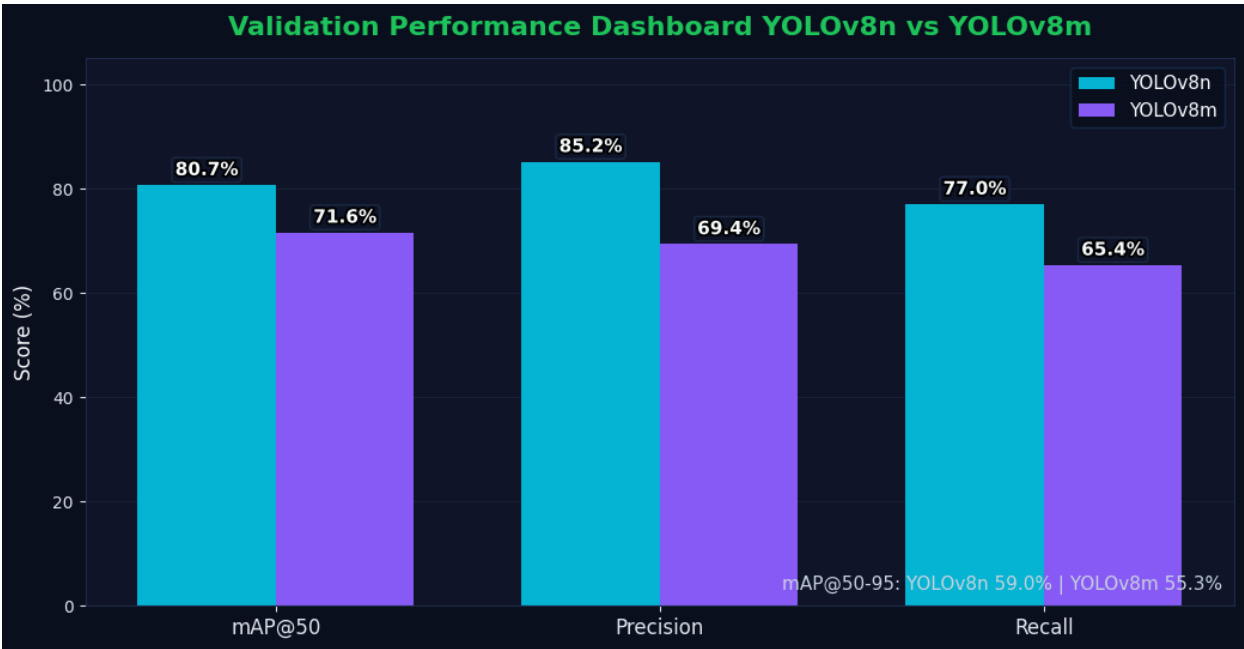
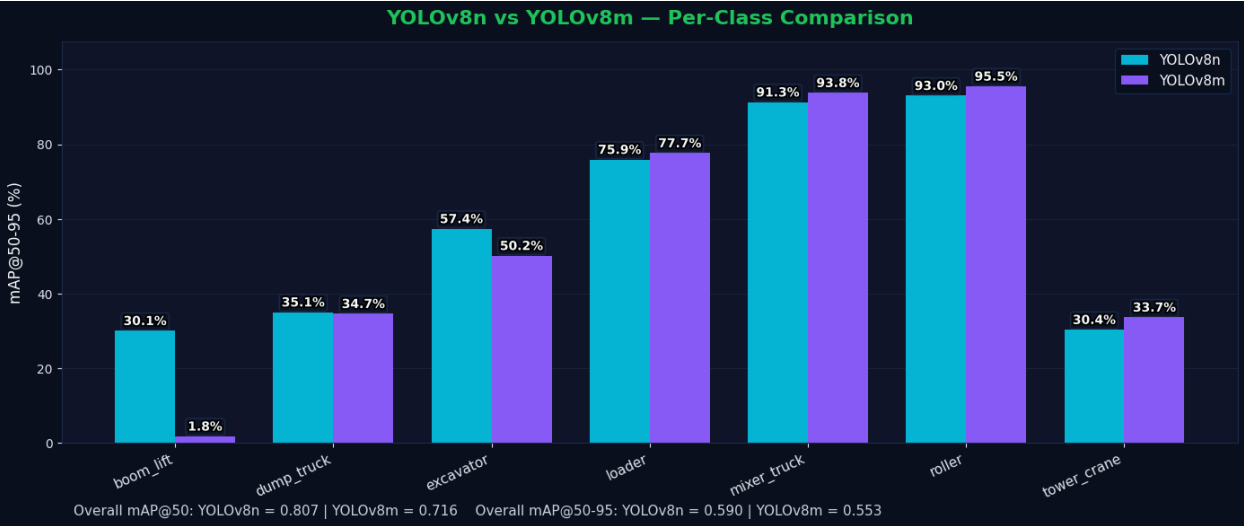
Rotation: Between -3° and +3°

Brightness: Between -15% and +15%

Exposure: Between -10% and +10%

Stage 3: Results, Experimentation & Model Selection

Experimentation with YOLOv8m vs YOLOv8s (Dataset v3, 581 imgs)- our initial experiment



Phase 1: The Data Starvation

Problem (Dataset v3, 581 imgs)

- Initial Testing: We benchmarked YOLOv8n against a heavier YOLOv8m architecture on our heavily constrained, 1,392 -image dataset.
- The Overfitting Reality: The medium architecture underperformed the nano model across all major metrics (71.6% vs 80.7% mAP@50).
- Early Conclusion: Increasing parameter count on a small dataset leads to severe overfitting. We deduced that radical data expansion was required before scaling model capacity.

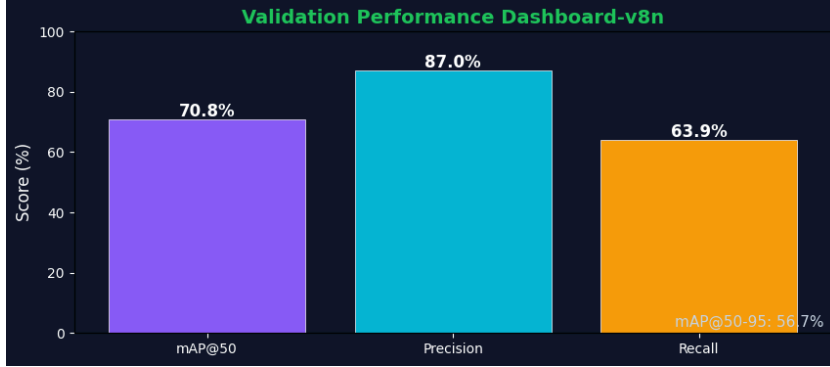
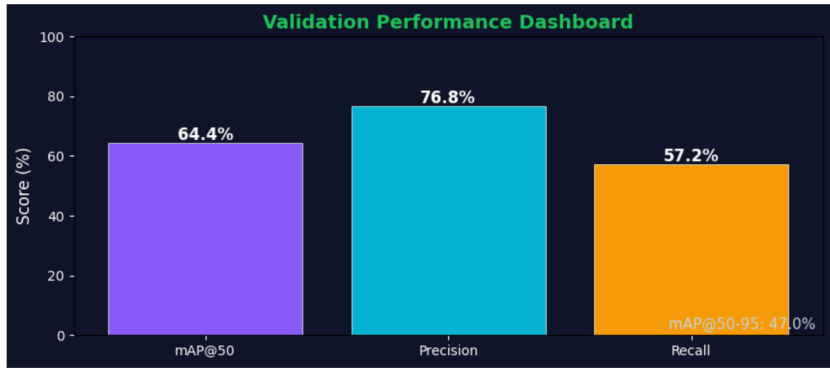
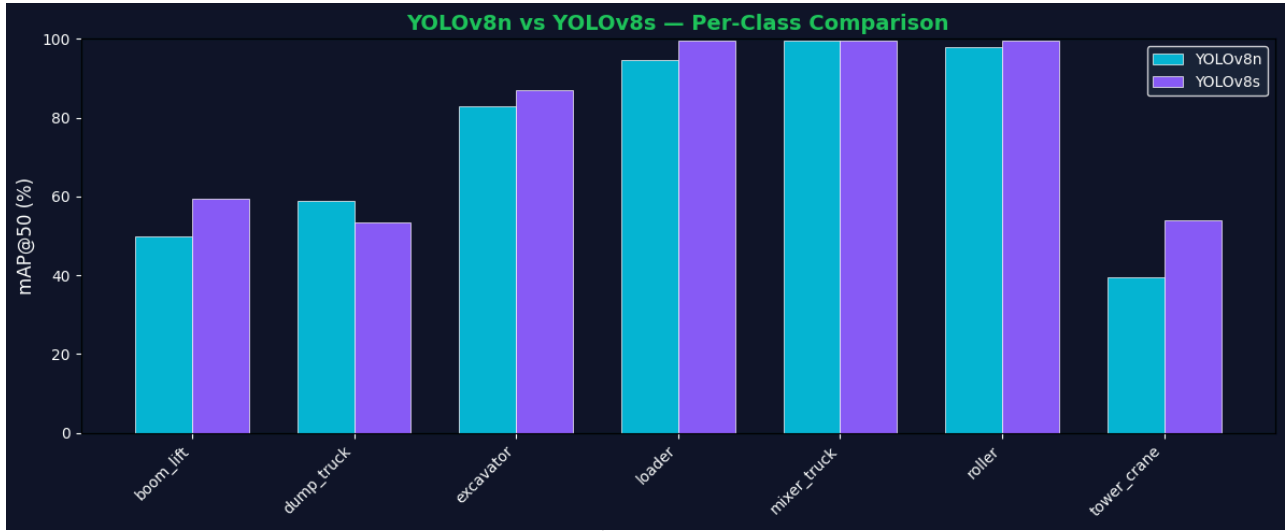
Figure 4: YOLOv8m underperforms YOLOv8n across most classes despite 3.7x more parameters -a clear sign of overfitting on a small dataset.

Phase 2: Data Scaling &

Architectural Success (Dataset v3, 581 imgs)

- The Solution: We scaled our dataset volume with an augmentation of x3, advancing to 1,392 images to feed heavier architectures.
- YOLOv8s delivers consistent improvements across all 7 classes, with the most significant gains in Precision (+10.2%) and mAP@50-95 (+9.7%). The heavier architecture extracts more nuanced features from the expanded dataset, particularly benefitting the previously underperforming classes. However, dump_truck and tower_crane still require targeted data collection to reach production-level accuracy.

Google Colab code results YOLOv8n vs YOLOv8s (Dataset v3, 581 imgs)- our selected model



Final Model Selection: YOLOv8s (Dataset v3)

- Conclusion: Our iterative testing proves that architectural capacity must scale synchronously with dataset volume. YOLOv8s trained on the expanded v3 dataset resolves our initial overfitting issues and provides the ultimate balance of high accuracy for live construction site deployment.

Stage 4: Visual Evidence of Learning

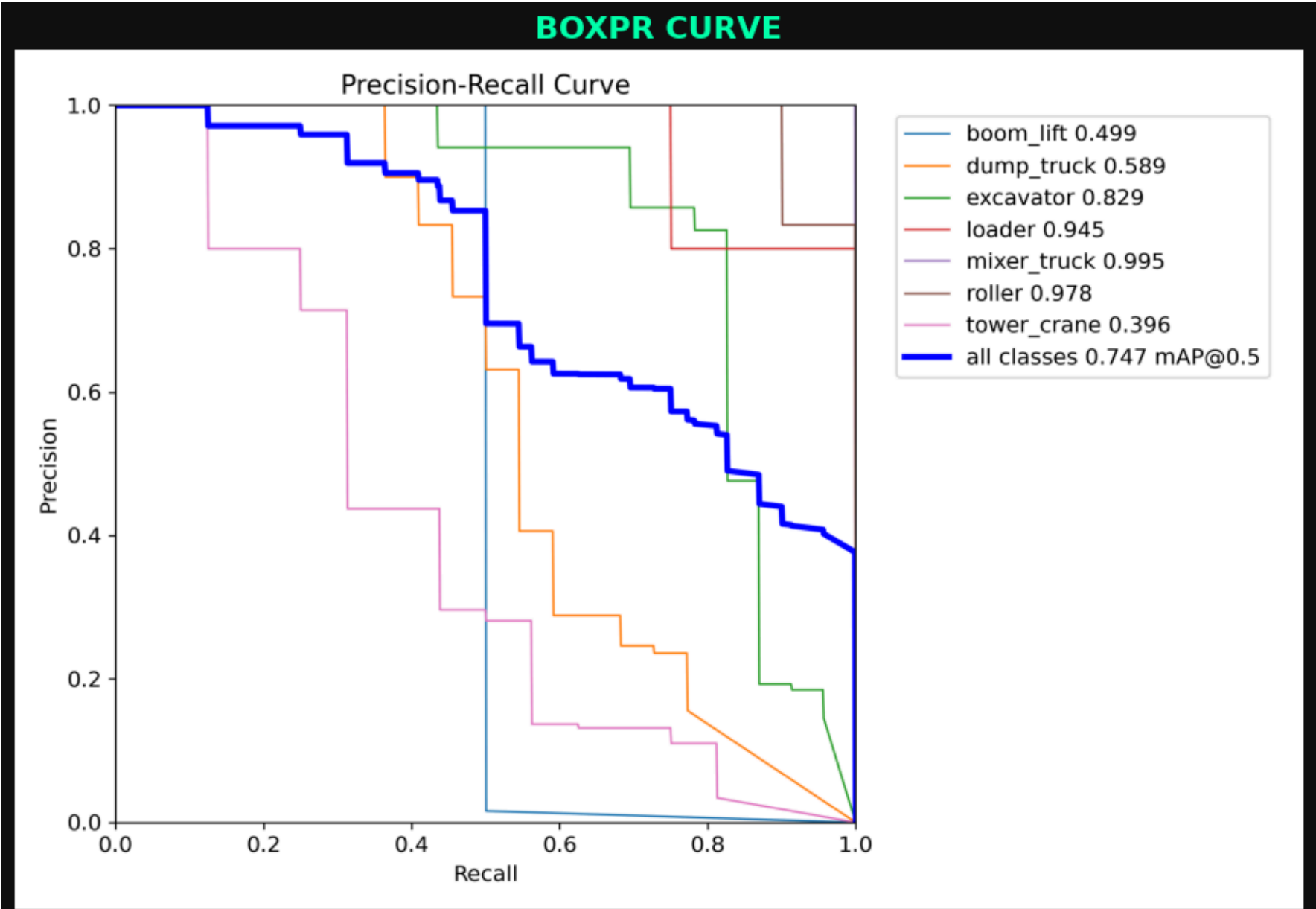


Figure 6: Precision-Recall (PR) Curve by Class

Precision-Recall Analytics

- **Exceptional Performance on Distinct Machinery:** The PR curve visually validates the model's extreme confidence on distinct, heavy vehicles. The system achieves near-perfect detection capabilities with a 0.995 mAP on mixer trucks, a 0.978 mAP on rollers, and a 0.945 mAP on loaders.
- **Overall System Health:** The updated model establishes a highly robust baseline, achieving a strong overall metric of 0.747 mAP@0.5 across all seven equipment classes.
- **Persistent Edge Cases:** The visualization honestly reflects our ongoing struggles with thin or complex structures. Tower cranes remain the most difficult class (0.396 mAP), dropping off sharply in recall, while boom lifts (0.499 mAP) exhibit a steep cliff, completely failing to recall instances beyond the 50% threshold.



Figure 7: Edge Inference on Unseen Validation Data (6x6 prediction grid)

Real-World Edge Inference

- **Robust Detection in Clutter:** Testing the model on unseen validation batches proves its field readiness. The system successfully generates tight bounding boxes despite severe site complexities.
- **Overcoming Variables:** The visual evidence confirms the YOLOv8 architecture successfully extracts machinery features regardless of scaffolding occlusion, varied lighting, and diverse camera angles.

Stage 5: Error Analysis & Iteration Plan

What Went Wrong

False Negatives (Most Critical):

- Dump Trucks (12 Missed): Vehicle colors and textures likely blend into the complex earthwork backgrounds.
- Tower Cranes (8 Missed): Thin lattice structures wash out against bright skies at the 512px training resolution.
- Excavators (4 Missed): Unusual machine poses (e.g., arm fully extended vs. tucked) were likely underrepresented in the training data.

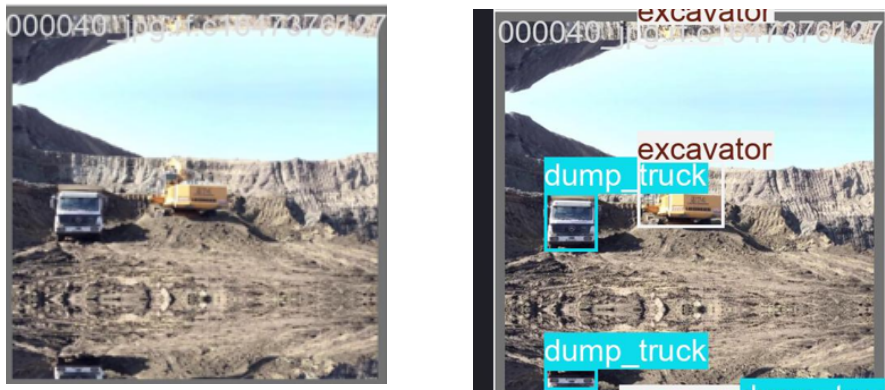


Figure 10: (Missed Detection): Comparison showing the model (left) failing to detect the dump truck present in the ground truth labels (right)

False Positives:

- Tower Cranes (7 Hallucinations): The model confuses other vertical site elements, like scaffolding or concrete pipes, for crane masts.
- Earthmovers (6 Total Phantoms): Deep shadows, complex dirt piles, or civilian vehicles mimic the blocky profiles of loaders, excavators, and dump trucks.
- Cross-Class Confusion (1 Instance): A raised dump truck bed was geometrically misidentified as an excavator arm.



Figure 9: Mixer truck chute misclassified as an excavator arm due to shape similarity.

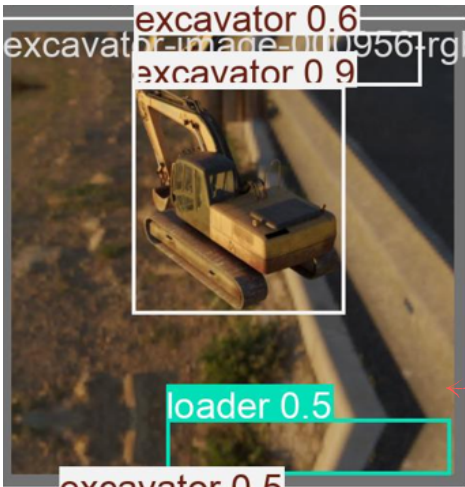


Figure 10: Model getting confused by shapes/colours

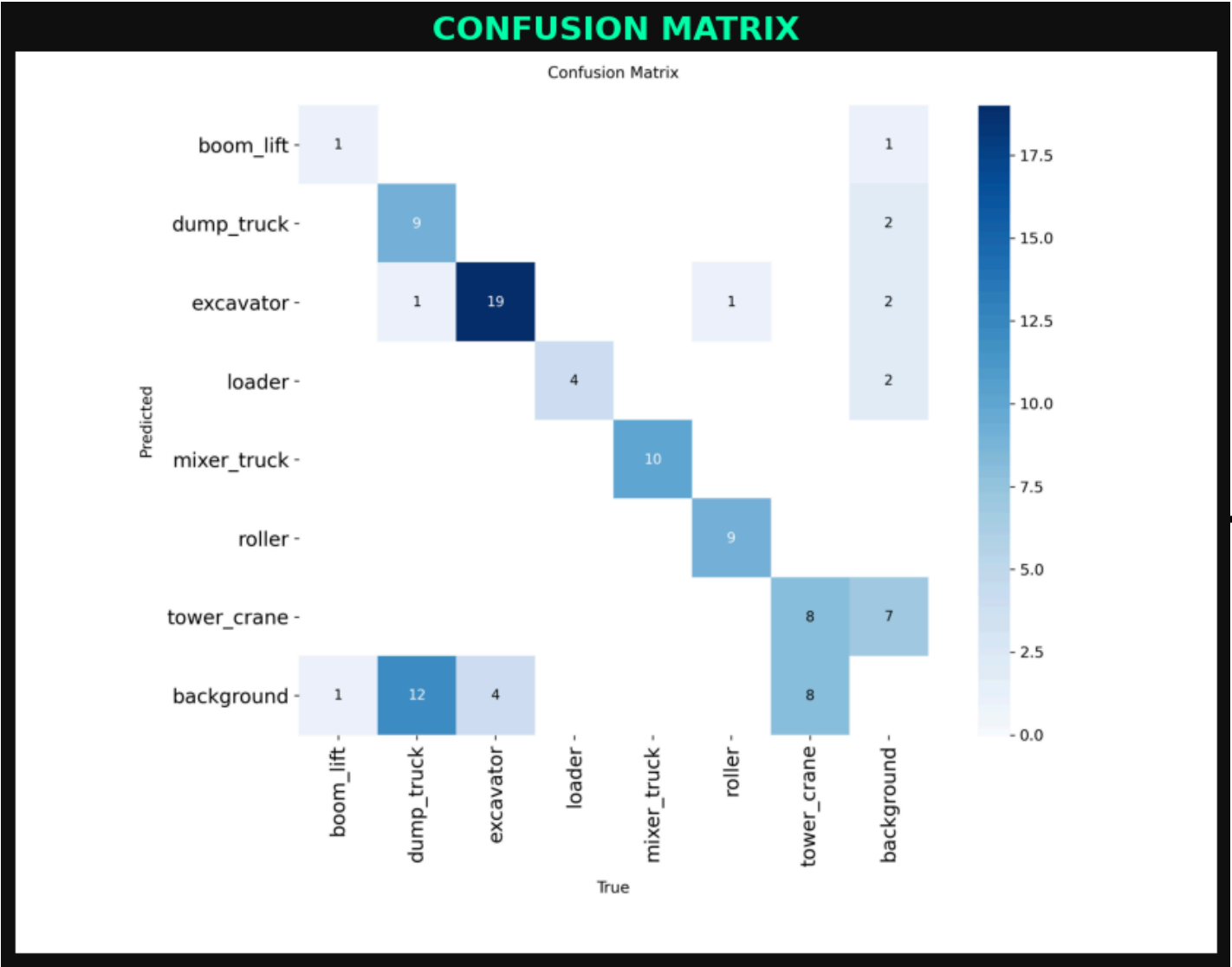


Figure 11: Confusion Matrix

Interpreting the Matrix

- Strong Class Separation: The diagonal squares confirm the model successfully distinguishes between different types of machinery. It very rarely confuses one active vehicle for another (e.g., only 1 dump truck was misclassified as an excavator).
- The False Negative Challenge (Missed Detections): The high values along the bottom "background" row highlight our primary issue. The model completely missed 12 dump trucks, 8 tower cranes, and 4 excavators, letting them blend into the site background.
- The False Positive Challenge (Phantom Detections): The rightmost column indicates the model occasionally "hallucinates" equipment where there is none. Notably, it incorrectly labeled the background as a tower crane 7 times, likely confusing thin vertical structures like scaffolding or poles for crane lattices.

How We Fix it (Iteration Plan)

- Targeted Expansion for Dump Trucks: To resolve the 12 missed dump truck detections, we will inject 300+ diverse, real-world images of dump trucks specifically positioned against complex earthwork backgrounds to prevent them from blending into the site terrain.
- Hard Negative Mining for Tower Cranes: The model missed 8 tower cranes and hallucinated 7 phantom cranes from the background. We will introduce images of empty scaffolding, vertical pipes, and bare skies without bounding boxes to explicitly teach the model what a crane is not.
- Resolution Scaling: To ensure the model stops losing the thin lattice structures of cranes to the sky, we will retrain the pipeline at a higher resolution (imgsz=640 or 1024) to preserve critical fine details.

Stage 6: Governance, Limitations & Licensing

Safety & Compliance



Figure 12: Typical site proximity. To comply with GDPR, worker faces must be dynamically blurred before this footage is stored or analyzed.

Safety Risk Assessment

- In AECO deployment, False Negatives are exponentially more dangerous than False Positives.
- A missed dump truck can cause safety alarms to fail, leading to fatal struck-by accidents.
- A phantom detection only causes a minor logging error.

Human-in-the-Loop Requirement

- This model is a monitoring aid only — it must NOT be used for automated safety-critical shutdowns without a human safety officer's verification.
- In safety-critical zones (e.g., tower crane swing radius), the model must be tuned for high Recall over Precision.

Privacy & Regulatory

- Face and license plate blurring required before any site deployment (GDPR compliance).
- EU AI Act: Construction safety monitoring may classify as "high-risk AI" under Annex III, requiring risk management documentation and human oversight provisions.

Operational Limits & Licensing

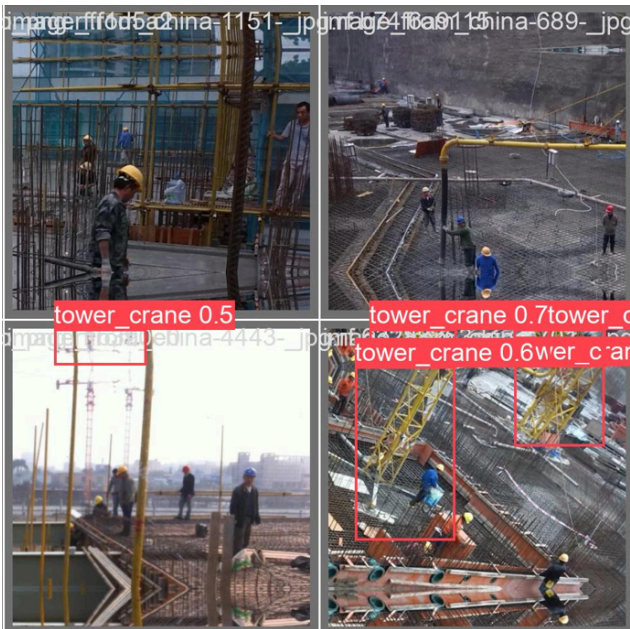


Figure 13: Operational Limitation. The model struggles with >100m distances and low-contrast lighting conditions

When NOT to Use This Model

- Not validated for nighttime or low-light conditions
- Cannot detect equipment >100m from camera (512px training resolution)
- Performance degrades in heavy dust, rain, or occlusion
- Not tested across different geographies or construction practices
- Training data biased toward daylight, clear weather conditions

Licensing

- Code: MIT License (open source)
- Dataset: Roboflow Universe (public, with attribution)
- Framework: Ultralytics AGPL-3.0 (commercial use requires Enterprise license)

Data Provenance

- 581 images from Roboflow Universe contributors
- Dataset may not represent all global construction environments
- Validate locally before deploying in new regions



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EQUIPMENT TRACKER

Thank You

